ORSKAR, Sweden

WMO: **024880**

Lat: **60.526N** Lon: **18.373E** Elev: **7** StdP: **101.25** Time Zone: **1.00 (EUC)** Period: **96-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	2	-10.1	-7.9	-13.1	1.2	-9.1	-11.1	1.4	-7.1	19.7	-0.1	17.6	0.1	6.1	220	0.826	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	7	5.1	24.4	18.3	22.7	17.7	21.2	17.1	19.6	22.6	18.6	21.5	17.7	20.5	5.3	120	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	18.5	13.4	21.0	17.4	12.5	20.0	16.5	11.7	19.3	55.8	22.8	52.7	21.6	49.9	20.5	22.6	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	17.7	15.7	14.0	DB	-12.2	27.3	4.2	1.9	-15.3	28.6	-17.7	29.8	-20.0	30.9	-23.1	32.3	(4)
(5)				WB	-12.4	20.3	3.7	1.5	-15.0	21.3	-17.2	22.1	-19.3	23.0	-22.0	24.0	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours		DBAvg	7.0	-1.2	-1.6	0.3	4.1	9.0	13.6	17.3	17.0	13.0	7.8	3.6	0.7	(6)
(7)			DBStd	7.31	3.17	3.67	3.21	3.02	3.23	3.02	2.70	2.42	2.50	2.76	2.73	3.16	(7)
(8)			HDD10.0	1777	348	323	302	178	58	4	0	0	4	80	191	288	(8)
(9)			HDD18.3	4171	607	557	560	427	289	144	54	57	161	328	441	547	(9)
(10)			CDD10.0	686	0	0	0	2	27	112	225	216	94	10	0	0	(10)
(11)			CDD18.3	37	0	0	0	0	0	2	21	14	1	0	0	0	(11)
(12)			CDH23.3	99	0	0	0	0	2	4	66	27	0	0	0	0	(12)
(13)			CDH26.7	8	0	0	0	0	0	0	7	1	0	0	0	0	(13)
(14)	Wind		WSAvg	7.1	8.0	7.6	7.4	6.6	6.0	6.0	5.7	6.2	7.0	7.9	7.9	8.6	(14)
(15)	Precipitation		PrecAvg	464	30	26	23	27	33	51	52	54	43	46	44	34	(15)
(16)			PrecMax	693	65	49	46	79	67	122	121	156	92	114	89	68	(16)
(17)			PrecMin	312	3	4	0	2	6	5	7	3	9	3	11	5	(17)
(18)			PrecStd	83	14	12	12	19	16	27	34	37	24	26	20	17	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		5.6	5.8	10.0	16.2	21.2	23.9	27.6	25.8	21.7	15.2	10.6	8.6	(19)
(20)			MCWB		4.5	4.2	5.9	9.8	13.9	16.4	19.2	18.6	16.4	12.9	9.7	7.2	(20)
(21)		2%	DB		4.6	4.6	7.4	12.9	18.0	21.8	25.1	23.8	19.3	13.6	9.2	6.8	(21)
(22)			MCWB		3.6	3.3	4.5	8.4	12.2	15.7	18.8	18.6	15.7	11.9	8.3	5.7	(22)
(23)		5%	DB		3.6	3.7	5.7	10.4	15.7	20.1	23.3	22.1	17.8	12.6	8.4	5.8	(23)
(24)			MCWB		2.8	2.5	3.5	6.9	11.0	15.0	18.2	17.9	14.9	11.1	7.5	4.9	(24)
(25)		10%	DB		2.9	2.8	4.4	8.5	13.8	18.4	21.7	20.8	16.5	11.6	7.5	4.7	(25)
(26)			MCWB		2.1	1.8	2.7	5.6	9.9	14.3	17.6	17.3	14.2	10.3	6.7	3.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		4.9	4.7	6.2	11.0	15.3	17.8	21.2	21.1	17.2	13.3	10.1	7.6	(27)
(28)			MCDB		5.4	5.6	9.3	14.7	20.1	21.8	24.3	23.6	19.9	14.5	10.5	8.5	(28)
(29)		2%	WB		3.8	3.6	4.8	8.7	12.8	16.7	20.0	19.5	16.1	12.2	8.6	5.9	(29)
(30)			MCDB		4.4	4.4	6.9	12.2	16.9	20.3	23.3	21.9	18.6	13.3	9.1	6.7	(30)
(31)		5%	WB		2.9	2.7	3.8	7.2	11.3	15.7	19.0	18.6	15.3	11.4	7.6	4.9	(31)
(32)			MCDB		3.5	3.5	5.4	9.9	14.8	18.9	22.3	21.1	17.3	12.5	8.2	5.6	(32)
(33)		10%	WB		2.1	1.9	2.9	5.9	10.4	14.6	18.0	17.7	14.5	10.4	6.8	3.9	(33)
(34)			MCDB		2.8	2.6	4.2	8.0	13.3	17.9	21.3	20.2	16.2	11.4	7.4	4.6	(34)
(35)	Mean Daily Temperature Range		MDBR		3.1	3.3	4.2	5.1	5.6	5.4	5.1	5.0	4.2	3.2	2.7	3.0	(35)
(36)		5% DB	MCDBR		3.2	3.8	6.3	8.8	9.4	8.6	7.9	7.2	6.3	4.3	3.5	3.3	(36)
(37)			MCWBR		3.1	3.2	4.1	5.1	5.0	4.4	3.6	3.5	3.5	3.2	3.4	3.2	(37)
(38)		5% WB	MCDBR		3.1	3.6	5.8	8.0	8.2	7.7	6.7	6.0	5.7	4.1	3.4	3.1	(38)
(39)			MCWBR		3.1	3.2	4.1	5.1	4.9	4.5	3.7	3.4	3.6	3.3	3.3	3.2	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.272	0.313	0.333	0.367	0.367	0.358	0.383	0.374	0.343	0.329	0.294	0.373	(40)
(41)		taud			2.202	2.284	2.349	2.348	2.418	2.484	2.434	2.453	2.523	2.423	2.232	2.633	(41)
(42)		Ebn at Noon			485	676	793	829	859	874	837	816	784	657	445	338	(42)
(43)		Edh at Noon			39	66	85	102	102	98	101	92	72	58	38	27	(43)
(44)	All-Sky Solar Radiation	RadAvg			0.24	0.81	2.32	3.94	5.31	5.95	5.53	4.29	2.66	1.15	0.32	0.15	(44)
(45)		RadStd			0.03	0.16	0.26	0.34	0.45	0.41	0.54	0.39	0.28	0.18	0.07	0.02	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	-80	N/A	N/A	N/A	(46)
(47)	Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-100	N/A	N/A	N/A	(47)

Nomenclature: See separate page